Aurchi Chowdhury

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EDUCATION

Bangladesh University of Engineering and Technology — **Nov 2022 – Present**
Bachelor of Engineering in Computer Science and Engineering | CGPA: 3.88/4.00 — *Dhaka, Bangladesh*

Viqarunnisa Noon School and College — **Jan 2009 – Dec 2021**
High School | GPA: 5.00/5.00 | Board Scholarship — *Dhaka, Bangladesh*

INTERESTS

Natural Language Processing · Multimodal AI · Computer Vision · Machine Learning · Mechanistic Interpretability

PUBLICATIONS

Causal Localization of the English Pivot in LLaVA: Mechanistic VLM Analysis and Training-Free Multilingual Steering — **Jul 2026**
1st Workshop on Multilinguality in the Era of Large Language Models (MeLLM), ACL 2026 | Archival — *San Diego, CA*

Where Do Affordances Crystallize? Video vs. Image Self-Supervised Encoders for Embodied Perception — **Jun 2026**
7th Annual Embodied AI (EAI) Workshop, CVPR 2026 | Non-archival Poster — *Denver, CO*

WORK EXPERIENCE

Undergraduate Research Intern — On-site
Progya: Adaptive Tutoring Platform | Dept. of CSE, BUET (UGC-funded) — **Nov 2025 – May 2026**

- Developing an adaptive tutoring platform using Bayesian Knowledge Tracing (BKT), building learning pathways aligned with the BCS exam syllabus.
- Supervisor: Md. Saifur Rahman, Md. Shariful Islam Bhuiyan

PROJECTS

MemoryLane | *PyTorch, ConvNeXt, U-Net, PatchGAN, VGG* — **Apr 2026**

- Framed grayscale-to-color mapping as a conditional image generation problem in the CIELab color space, building a ConvNeXt-Base encoder – U-Net decoder architecture.
- Incorporated a PatchGAN discriminator and mitigated the gray-mean saturation bias endemic to deep colorization models via colorfulness-filtered training on ∼518K Places365 images.
- Designed a composite perceptual + adversarial loss (VGG feature matching + PatchGAN) to balance global structure with local texture fidelity.
- Surpassed ground-truth colorfulness (40.9 vs. 36.9) and generalized to real, out-of-distribution family photographs.

Scramble | *ATMega32, SG90 servo motors, HC-05 Bluetooth module, Kotlin* — **Jul 2025**

- Developed a Fully Automated Rubik's Cube Solver capable of generating solutions in a maximum of 20 moves.

Project Demonstration

On the Road | *Node, Express, React, PostgreSQL* — **Mar 2024**

- Built a full-stack travel planning platform covering accommodations, dining, attractions, and tour guides.

ACHIEVEMENTS

- **1st Runner Up, ResearchSpark 3 Minute Thesis Competition | BUET** — **Apr 2026**
- **1st Runner Up, HULT Prize at BUET** — **Feb 2026**
- **2nd Runners Up, Research Poster Presentation | BUET CSE Fest 2024** — **Oct 2024**